Grokking Under a Fixed Budget

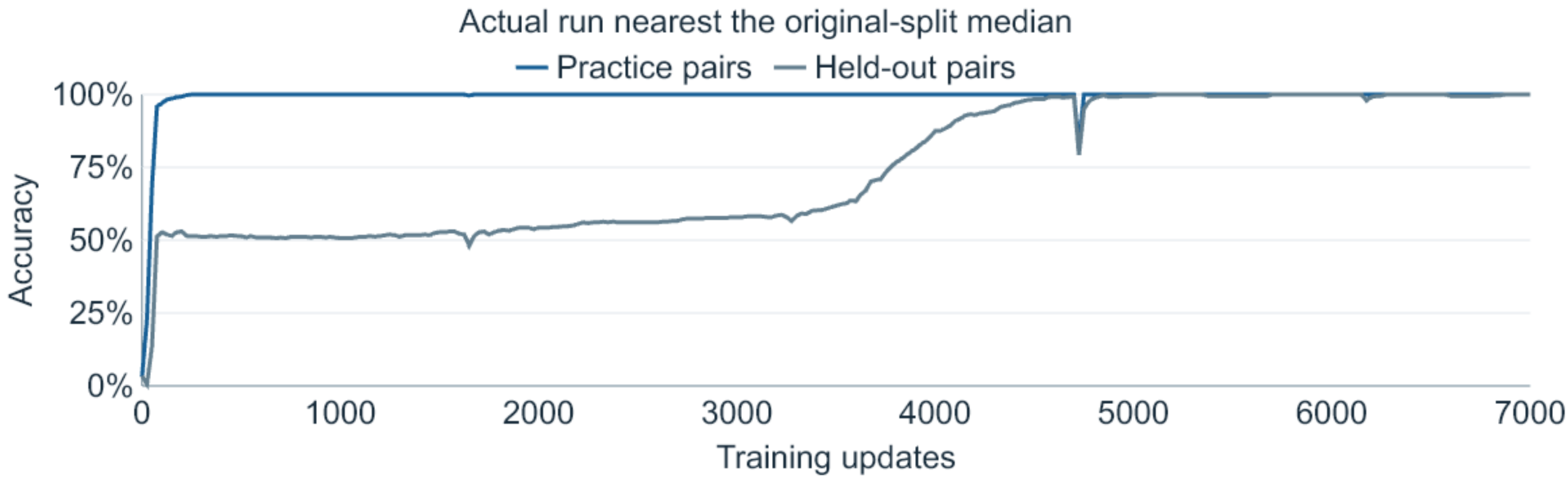
How should we spend 8,000 training steps?



Same total budget. Fresh starting weights for each run.

Name: [Your name]
Registration: [Your registration number]

Grokking: practice improves first



Memorization first

Generalization later

Keep the remainder

30 + 4 = 34 → 34 mod 31 = 3

The task and the dataset

961 ordered pairs: learn (a + b) mod 31



For learning

For evaluation

S1 **S2** **S3**

Three fixed partitions. Independent starting seeds within each.

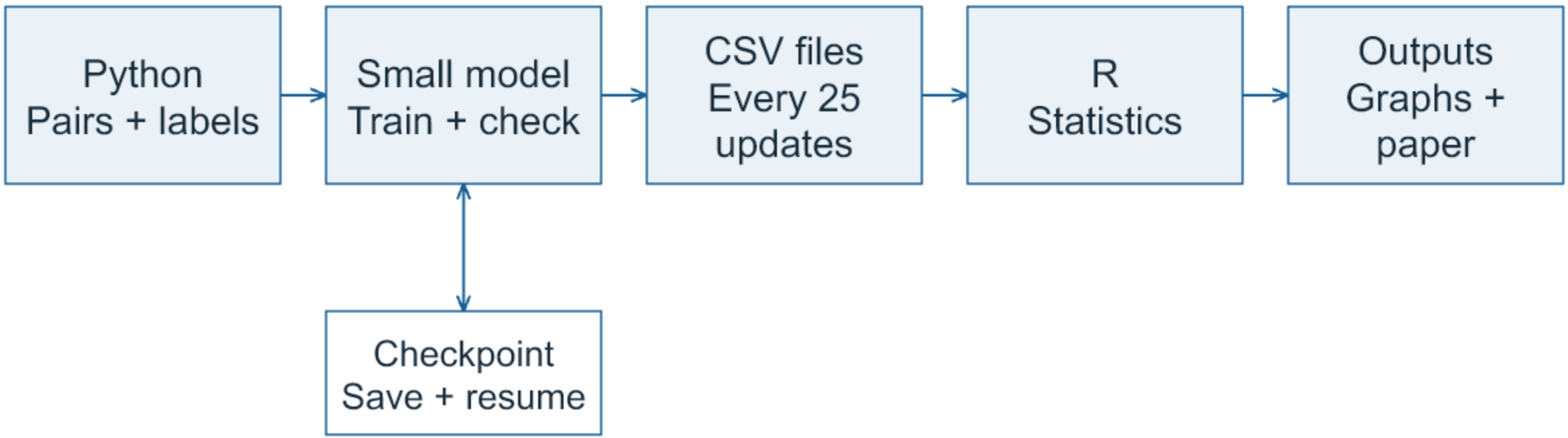
90 calibration observations

20 reused + 70 new calibration runs

540 fresh budget attempts

How the experiment works

Fixed model: 36,992 parameters



1,330 new runs

7.5 million logical updates

How we confirm success

Every good check: practice ≥99% and held-out ≥95%

First confirmation T



3 checks

21 checks

500-update window



Same scale: updates since the first qualifying check.

Count the final check. It must fit inside the allocated run budget.

One budget, different allocations

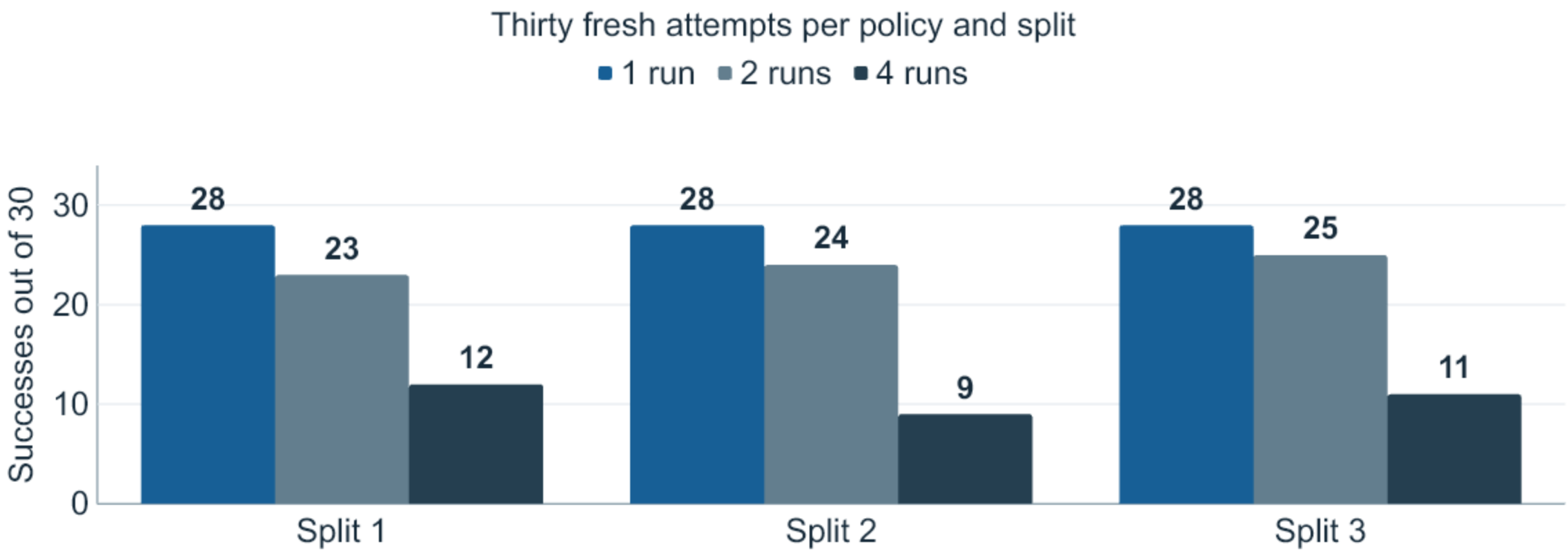
Total B = 8,000 updates in every row



$P(\text{at least one success}) = 1 - [1 - F(B/k)]^k$

F(t): chance one run succeeds by t. k: independent starts.

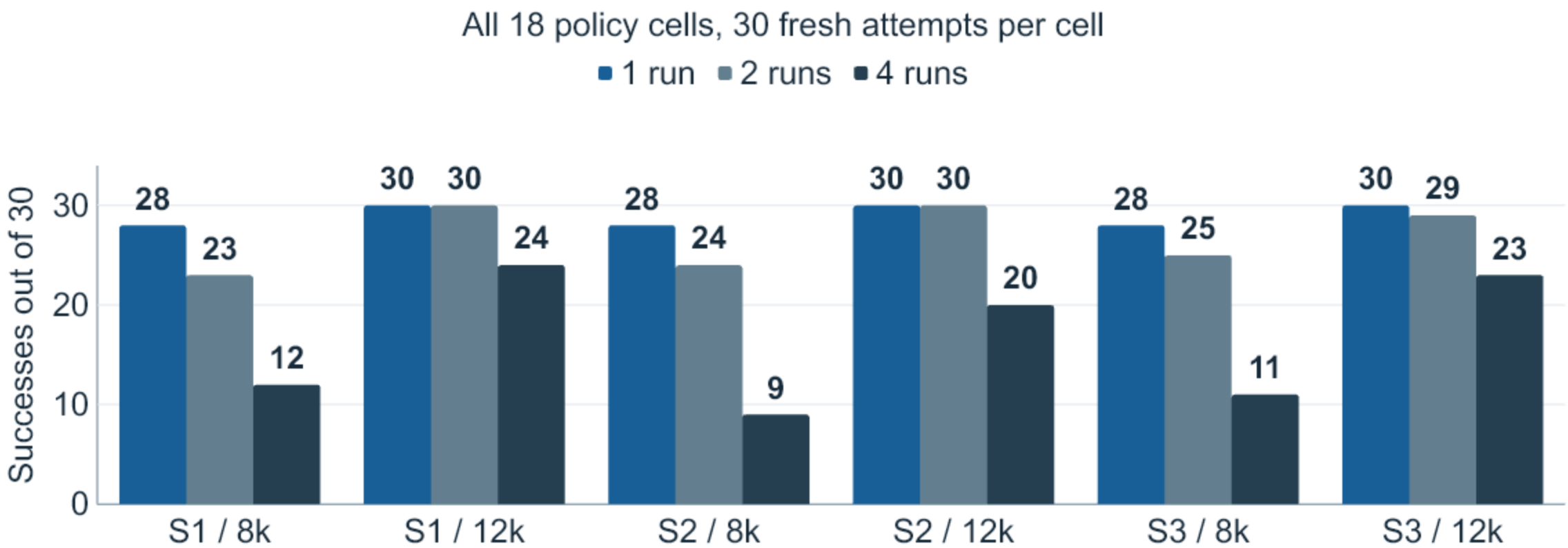
Results at 8,000 total updates



Four starts underperform one run on all three splits.

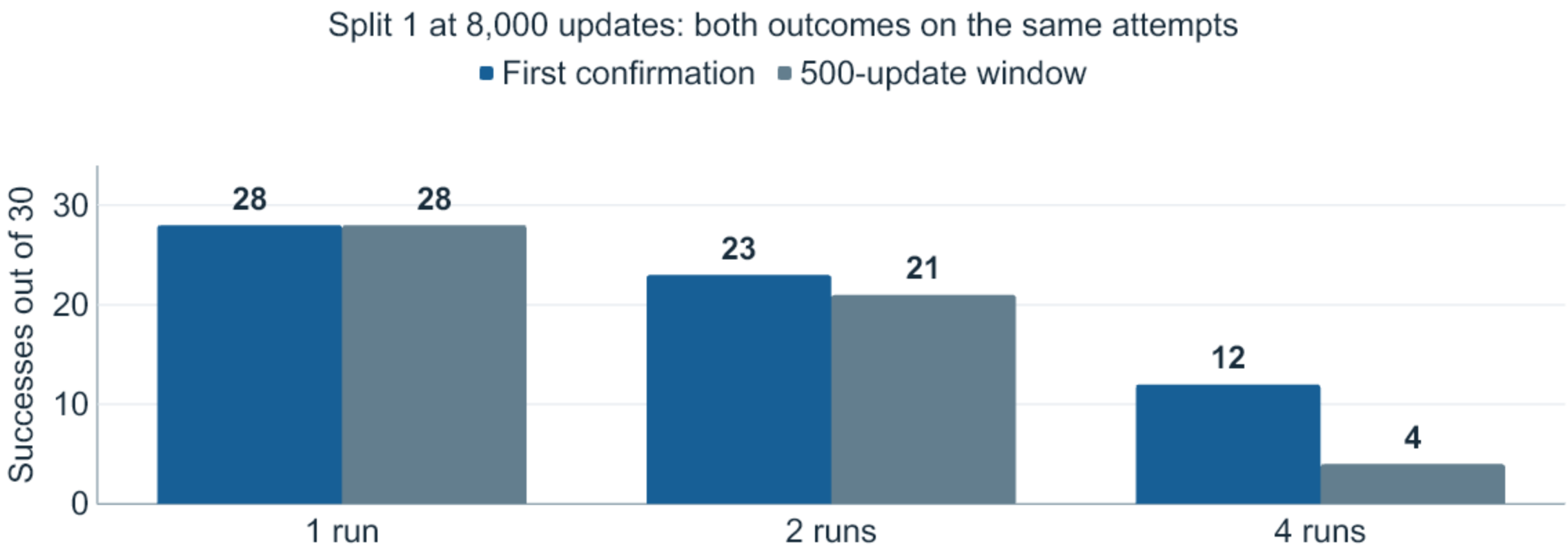
All three survive Holm correction. Two-run comparisons remain inconclusive.

All splits and both budgets



At 12k, only split 2 detects a four-versus-one difference after correction.

First confirmation and a longer window

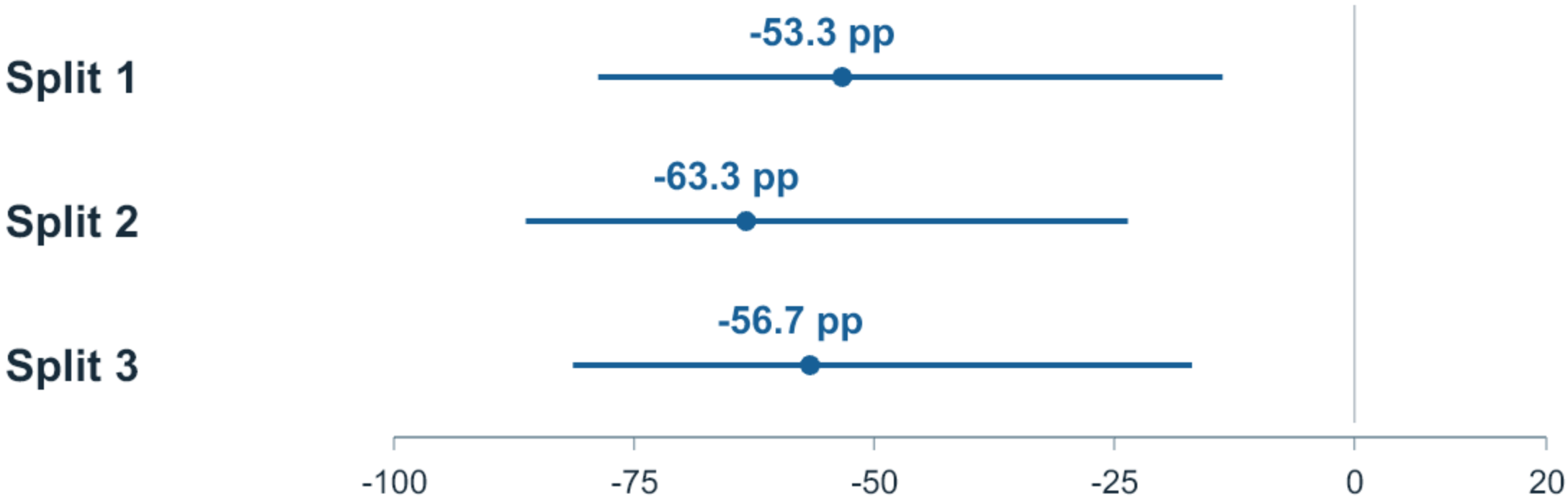


Four starts: 12 first confirmations, only 4 longer-window confirmations.

Finite window. Same budget. Future work: more tasks, splits and longer checks.

One long run was more reliable at 8k

Chance difference (percentage points): 4 runs minus 1



Negative differences favour one long run.

Conservative pointwise 95% intervals. Two-run comparisons remain inconclusive.